Four comparisons, one pipeline.

A five-stage football commentary system, measured against human commentary, live constraints, event annotations and production cost. The results are mixed in a useful way: the expensive parts are cheap, the goal calls work, and the parts that still fail are specific enough to name.

Aryaman · football-commentator · commit 7f1a826 · 64.5-min ARG–FRA second half

38/40
v4 lines accepted by the grounding validator

2/2
goals found, with precision and recall 1.00

0.40
overall event recall at ±1 min

$0.39
cloud compute for a 64.5-min match

TEST SETUP

What was actually measured

All numbers below come from the corrected full-match run and the held-out evaluation set. The reference moments are the ten curated moments from the earlier reel, not an exhaustive independent annotation; that distinction is kept explicit throughout.

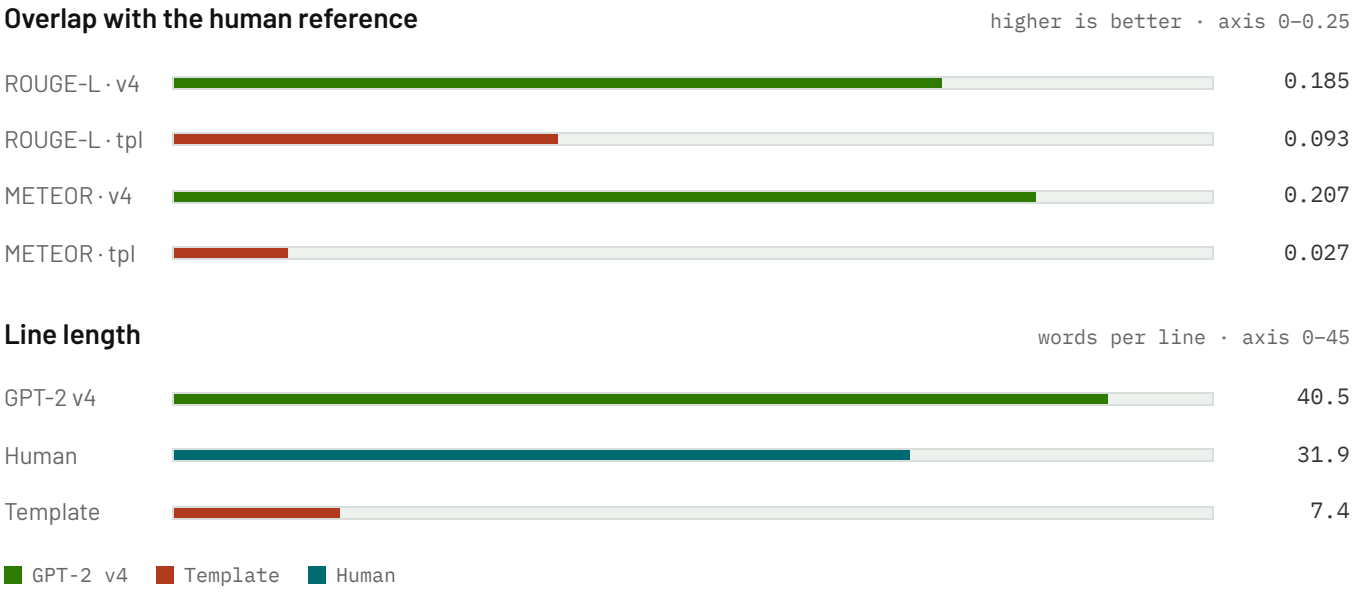
PIPELINE	
Perception	YOLO26n + ByteTrack, jersey clustering, possession, bounded ball interpolation.
Event layer	Score-bug anchors, motion-gated window scan, ResNet-18 classifier, crowd-audio rise.
Facts	Deterministic score, minute, ball zone, possession and event-team sheet.
Language	GPT-2 LoRA v4, validator, grounded template fallback.
Speech	Fish Audio s2.1-pro-free, event-conditioned pacing, ffmpeg mix.

MODELS AND RUN	
YOLO26n	2.39 M parameters, 640 px, 50 epochs, 2.70 h on an RTX 2050 4 GB.
GPT-2 v4	LoRA r=32, 3 epochs, 2,481 steps, final loss 0.779.
Full match	64.5 min, 43 commentary lines, 43/43 voiced, 19.4 min wall time.
Evaluation	40 held-out SoccerReplay prompts; 10 curated event moments; one 5-min causal chunk.
Hardware	One consumer GPU, one laptop, no cloud during inference.

COMPARISON 1

AI vs human commentary

GPT-2 v4 is closer to human commentary than the template baseline on ROUGE-L and METEOR, and it writes human-length lines. It is still not factually reliable: the examples below show invented detail, wrong teams and a missed goal.



Exact scores

n=40 held-out prompts

METRIC	GPT-2 V4	TEMPLATE
BLEU-1	0.0300	0.0103
BLEU-2	0.0014	0.0020
BLEU-4	0.000304	0.001038
ROUGE-L	0.1845	0.0931
METEOR	0.2066	0.0269
Average words	40.5	7.4
Grounding accepted	38/40	–

BLEU-4 is near zero for both systems. ROUGE-L and METEOR favour GPT-2; overlap metrics do not measure whether a line is true. The human reference averages 31.9 words, so GPT-2 is slightly longer than the humans, not shorter.

Three records, three different failures

Quoted verbatim from the 40 held-out records. Annotation shows the difference that matters, not a cherry-picked "best" line.

CLOSEST TO HUMAN · pass / West Ham

Human

Wonderful movement by West Ham. The players are holding possession, mostly exchanging one-touch passes. They control the pace of the game now. The midfielder attempts to feed the defender with a chip pass, but the goalkeeper reacts well and comes off his line to intercept.

GPT-2 v4

West Ham are showcasing great team work by exchanging some short passes and one-two passing balls with one another to create new attacking opportunities. The midfielder is unlucky enough not to shoot from the edge of the box as his effort flies just wide of the right post.

Template

Patient stuff from West Ham, switching the play smartly.

team correct | length 48 vs 56 | invented shot

WRONG TEAM · corner / Eintracht Frankfurt

Human

The forward strikes the ball towards goal, but it fails to reach the target after being well blocked by one of the defenders. Corner kick. Eintracht Frankfurt will have an opportunity.

GPT-2 v4

Napoli failed to take advantage of the corner as their opponents were alert and averted the threat at the last moment! The ball goes out of play. Neat result from the forward's attempted effort with a cross into the box resulting in a dangerous clearance.

Template

The delivery comes in for Napoli!

wrong team | corner topic

MISSED GOAL · goal / B. Monchengladbach

Human

The goal stands! That took a long time for the referee to decide. The relieved B. Monchengladbach players can finally celebrate.

GPT-2 v4

The forward races towards goal but the defender gets back well to make a challenge. One of the defending players reacts superbly to get his teammates down and intercepts one. A promising stop for Inter! Corner kick from long range.

Template

GOAL! Inter make it count — sheer precision at a vital time!

AI misses the goal | template calls it | both wrong team

> **Verdict.** GPT-2 is the better surface realiser: longer, more varied, closer to human overlap. It is not the better fact layer. The validator accepts 95% of lines, but acceptance means "passes the grounding checks," not "is correct." The template is shorter and safer, yet its team attribution fails in the same record, which points to the shared feature layer as well as the language model.

COMPARISON 2

Live vs pre-recorded

The full-context run gets both goals right. A 5-minute causal chunk is faster to first line, but it attributes both goals to the home side and invents an earlier goal. The current system is

pre-recorded only; chunking alone does not make it live.

PRE-RECORDED · FULL CONTEXT · CORRECT

[44:18] **(goal)** GOAL! France make it count — sheer precision at a vital time!

[45:27] **(goal)** IT'S A GOAL! Clinical from France — the keeper had no chance!

2/2 goals | team correct | first line at 16.6 min

LIVE-LIKE · 5-MIN CHUNK · WRONG

[02:01] **(goal)** GOAL! Argentina make it count — sheer precision at a vital time!

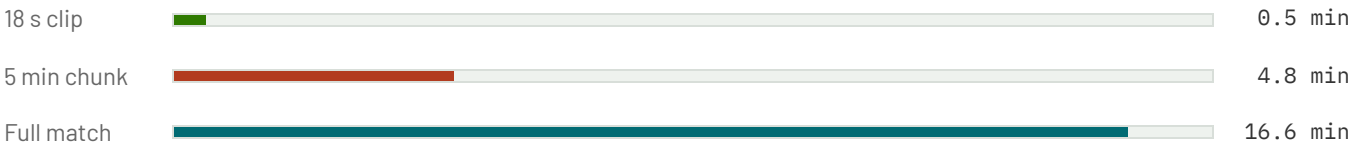
[04:05] **(goal)** IT'S A GOAL! Clinical from Argentina — the keeper had no chance!

wrong side | false early goal | first line at 4.8 min

Ground truth from the score-bug reader: the two anchors in the chunk are `away 44:33` and `away 45:42`; the broadcast score flips 2–1 to 2–2. The chunk's generated lines still say Argentina. The side is lost between the anchor and the realised line, and the event classifier also promotes an earlier passage to a goal. This is why "faster" is not the same as "live."

Time to first commentary line

lower is better · axis 0–18 min



The chunk is faster because it sees less. The full run waits for the whole match before writing and voicing; the chunk still processes a whole 5-minute block before the first line. Neither is a streaming system with bounded latency.

Wall time versus real time

1.00× = real time

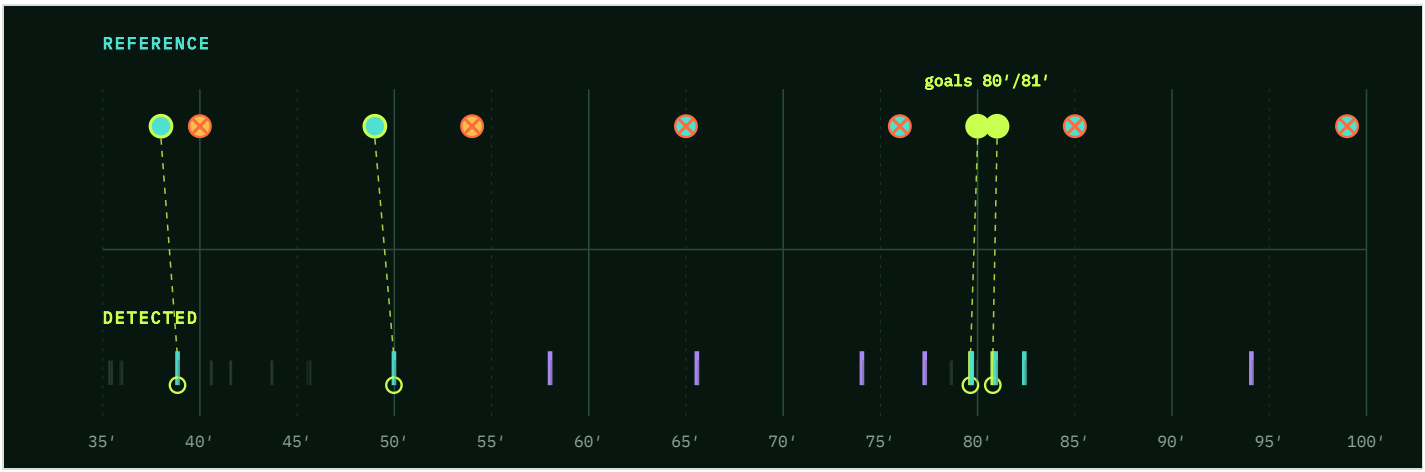
MODE	INPUT	WALL	RATIO
18 s clip	18 s	39.8 s	2.21×
5-min chunk	300 s	311.8 s	1.04×
Full match	3,871 s	1,165 s	0.30×

> **Verdict.** The full-match run is faster than real time in aggregate but not live: 16.6 minutes of latency before the first line. The causal chunk is nearly real time but factually wrong. A live mode needs streaming perception, incremental TTS and explicit side propagation, not just shorter inputs.

COMPARISON 3

Event spotting

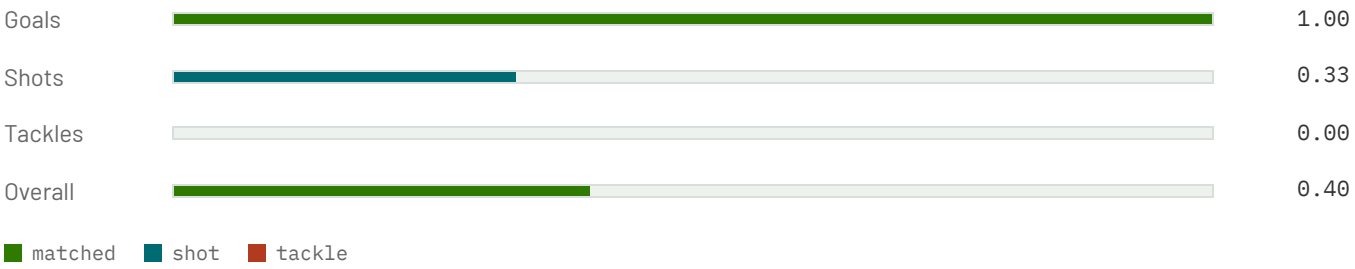
Goals are the strong case: the score bug gives a precise anchor and both goals are found. Shots and tackles are the weak case: the window classifier finds only two of six shots and none of the two tackles within a minute.



Top: the ten curated reference moments. **Bottom:** the 38 event lines from the full-match run. Faint ticks are passes and dead time; the stronger ticks are goals, shots, corners and tackles. Solid connectors mark the four matches within ±1 min; coral X marks a missed reference event.

Recall at ±1 min

n=10 reference moments



Goal precision and recall against the known 80' and 81' goals are both **1.00**. Timing MAE for matched events is 36.2 s; goals are 13–22 s early, shots 51–59 s late.

What the pipeline called

38 event lines

LABEL	COUNT
pass	19
dead time	7
shot	5
corner	5
goal	2

The pipeline over-produces pass and dead-time lines and under-produces the reference shot/tackle classes. This is an event-classification problem, not a language problem.



The 81' France goal clip used for the score-bug test. The broadcast score flips from 2–1 to 2–2; the reader now infers the starting score and anchors the away goal at 5 s. Contact sheet: match frames at 0, 3, 6, 9, 12 and 15 s.

Reference vs detected

±1 min tolerance

MIN	REFERENCE	DETECTED	RESULT
38	shot	38.85 shot	matched +50.8 s
40	tackle	—	missed
49	shot	49.98 shot	matched +58.8 s
54	tackle	—	missed
65	shot	65.56 corner	missed
76	shot	—	missed
80	goal	79.63 goal	matched −22.2 s
81	goal	80.78 goal	matched −13.2 s
85	shot	—	missed
99	shot	—	missed

Reference moments are curated, not exhaustive, so this is a recall/consistency check; precision for shots/tackles/corners is not measured here.

Object detection is not event spotting separate metric

MODEL	PARAMS	EPOCHS	MAP50	MAP50–95
YOLO26n	2.39 M	50	0.805	0.481
YOLOv5lu	53 M	33	0.795	0.459



YOLO26n is fractionally ahead, but the runs are not equal-budget: 50 vs 33 epochs, 2.39 M vs 53 M parameters. Per-class YOLO26n mAP50–95: Player 0.621, Ball 0.291, Referee 0.530. The low Ball score is why the ball path is interpolated rather than trusted.

Verdict. The score-bug goal path is reliable enough to build on. The general event classifier is not: it finds one third of reference shots, no tackles, and produces almost twice as many pass/dead-time lines as event lines. Report goals and object detection separately; do not turn detection mAP into an event-spotting claim.

COMPARISON 4

Cost

On this laptop, a 64.5-minute match costs about one cent of electricity; on a cloud A10G it is under forty cents. Human commentary, even at a modest illustrative rate, is hundreds of times more expensive. The barrier is quality and latency, not compute.

ITEM	LOCAL RTX 2050	CLOUD G5.2XLARGE	HUMAN TEAM
Inference per match	$0.010	$0.392	$300 live / $400 pre-recorded
Training (both models)	$0.109	$4.40	—
Training time	3.64 h	3.64 h	—
TTS	43 calls	43 calls	—
Speech	412.2 s	412.2 s	—

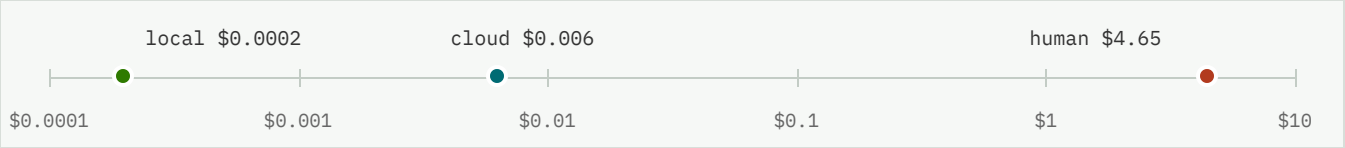
400–1,500×

cheaper than the illustrative human team on cloud compute, depending on whether the human rate is $25 or $100 per hour.

Assumptions: cloud $1.21/h; local 100 W at $0.30/kWh; two-person human team at $50/h for 3 h live / 4 h pre-recorded. Human rates are illustrative, not a quote. Fish Audio is on the free tier; paid TTS, storage and egress are excluded.

Cost per match minute

log scale · lower is better



local cloud human

A log scale is necessary here: local is 23,000× cheaper than the human scenario per minute. The numbers are the measured/assumed values from the table.

> **Verdict.** At current quality, the system is economically credible as a pre-recorded or offline tool. A live product would add streaming infrastructure and still need the quality fixes; those costs are not in this table.

DISCUSSION

What the numbers support

The honest claim is narrower than "AI commentary." It is a cheap, auditable pre-recorded pipeline whose goals are reliable and whose general event layer is not yet good enough for broadcast output.

HOLDS	
Goals	2/2, precision and recall 1.00 via the score bug.
Cost	$0.39 cloud, $0.01 local per match.
Auditability	Facts, validator and ClearML runs are inspectable.
End-to-end	64.5 min in, 64.5 min voiced out, 43/43 lines.

DOES NOT HOLD	
Live	16.6 min to first line; chunking is faster but wrong.
Events	0.40 overall recall; no tackles; two of six shots.
Teams	Wrong side survives both AI and template paths.
Language	Invented detail, repetition and malformed names.

WHY	
Score bug	High precision, but the side must be carried into the line.
Classifier	ResNet windows over-produce pass/dead time; rare events are missed.
Chunking	Short context loses the match story and makes false goals easier.
Validator	Checks words, not truth; it accepts 95% of lines.

> **Paper claim.** The pipeline is a working, low-cost, auditable pre-recorded commentary system. Its strongest contribution is the separation of deterministic facts from language and the score-bug goal path; its weakest links are general event spotting, team-side propagation and live latency. The paper should report those weaknesses as findings, not hide them.

Reproduce

```
python scripts/compare_commentary.py --n 40 --device cuda
python scripts/evaluate_event_spotting.py
python scripts/compile_experiments.py
```

AI vs human data	output/comparison/commentary_comparison.json · docs/figures/commentary_metrics.png
Event data	output/experiments/event_spotting.json · output/experiments/full_match_job.json
Cost data	output/experiments/experiment_results.json · docs/experiments_results.md

Full-match output	output/1793056646_3 - 3 ARG vs FRA - Second-Half - Bein_0s/final_commentary.mp4

Matchcall · football-commentator · experiment report generated 27 October 2026. The human cost figures are illustrative assumptions, not vendor quotes. Reference moments are curated, not exhaustive; object-detection mAP is not event-spotting accuracy.